

Chance, made permanent

Quantum-Film, film grain laid by a quantum circuit on Moth's Atlas and on IBM's quantum processors, developed on pinned arithmetic, and fixed bit for bit

Quantum-Film makes film grain with a quantum computer and then makes it permanent. A 16-qubit circuit prepares five fermions on a four-by-four tile of sites, and measuring every qubit places five crystals that keep their distance from one another, because fermions obey Pauli exclusion. The circuit ran 24,576 times on Moth's Atlas emulator and 761,856 times on three of IBM's quantum processors, and every shot was fixed as a record whose commitment was made before its result was read. Stacked twenty-nine layers deep, the shots coat a sheet of film at the density of TRI-X, developed and printed on pinned arithmetic, so the print is the same bits on every machine. The limits come first: the laws are small enough to simulate classically, which is what makes them checkable; the processors ran for one day, on a free plan; and the prints are fractions of a millimetre of film. What the grain shows, between a tile's scale and the Fermi length, is a structure no randomly placed emulsion gives. What the processors show is the law surviving their noise: 4.3% to 5.2% of their five-crystal shots landed where the law forbids, against 31.1% for crystals placed at random, and the ones that did were clumps.

1 What it is

Quantum-Film is a set of film stocks whose crystals are laid by quantum circuits, and a fixer that makes every print of a roll the same, bit for bit, on every machine [1]. A roll of film is chance made permanent: silver halide crystals land where they land, and the fixer bath locks the image in. This project makes both halves literal. A quantum measurement lays the crystals, under the Born rule, which no machine can repeat; the fixer writes the roll down as integers with its provenance; and everything after that is arithmetic that gets the same answer everywhere.

The limits come first, because they decide what the work is worth. At these sizes the quantum laws are classically simulable, and that is exactly what makes them checkable, so

nothing here is a quantum advantage: “quantum” means that the crystals were laid by coherent interference and checked against the exact law. The circuit ran first on the emulator of Moth’s Atlas platform, because the account used there had no processor access [2], and then for one day on three of IBM’s quantum processors, on a free plan [3]: enough to measure what noise does to the law, not how a processor varies from day to day. And the prints are small: the largest is 284 columns of crystals on a side, about a third of a millimetre of film, seen crystal by crystal.

1.1 What exists

Everything described here was built between 25 September and 5 October 2026, for Moth Hack 2026, in the open under the MIT licence, and it all runs.

- The definition of correct is an exact sampler in Python, at 256 bits with uniforms taken from SHA-256, which refuses any draw that rounding could decide.
- A hardware-shaped circuit of 51 Givens rotations and 102 CNOTs lays the shelf’s Pauli law on 16 qubits. It ran on Atlas’s tomography-api-v2 engine in six jobs of 4,096 shots, and all 24,576 shots were fixed as records.
- The same circuit ran on IBM’s `ibm_kingston`, `ibm_marrakesh` and `ibm_fez` in 65 jobs, every measure pre-registered and every job’s line pushed to GitHub before its result was read. Each device’s shots laid a sheet of film of its own.
- A second sampler, in binary64 on the pinned arithmetic of `cft-fp256`, lays the exact sampler’s roll by construction.
- The develop seam turns rolls into a sheet of film and develops it through atlas-film, the film model this project borrows everything after coating from. Five first prints came out of it, one of them laid entirely by Atlas.
- A runner of 18 stages checks all of it, every gate watched to fail on a planted fault, and an append-only ledger records every run that established something.

1.2 How to read this paper

The next section says why grain is worth this much trouble. The three after it describe the law and the one sampler that decides it, the circuit that lays it, and what happened on Atlas. Then the fixer, the second sampler, the step from rolls to film, and what the grain shows; then the circuit on IBM’s processors, and what their noise did to the law. The last sections say how it is all checked, what it is not and what comes next, and how it was made.

Numbers in this paper are of two kinds. Some were recomputed from the repository while the paper was built: the counts of shots and layouts, the circuit’s shape, the prints’ figures, the hardware’s measures, the stages and the history. The rest are measurements made on a day and quoted from the document that records them, usually the validation ledger, because a measurement is a fact about a run, and the ledger that records it is never edited to agree with a later one.

2 Grain is where the crystals landed

A photographic emulsion is gelatin holding silver halide crystals. Light makes some of them developable, the developer turns those to silver, and the fixer washes the rest away. What is left is the image, and its texture is grain. Grain is not noise laid over a picture. It is the picture's own material, counted: where the crystals happened to land, how many of them light reached, and how they happened to clump.

That makes the placement of the crystals the thing to change. A classical emulsion places its crystals very nearly at random, and the statistics that can tell one kind of grain from another read the placement directly: the pair correlation of crystal centres, the structure factor $S(k)$, which is the noise power of the crystal count at spatial frequency k , measured per crystal, and Selwyn's law of how the fluctuation falls as the measuring aperture grows. A single granularity reading cannot, because it folds density, grain size and correlation into one number. For crystals placed at random the structure factor is about 1 at every frequency.

Free fermions are not placed at random. The positions of fermions filling a set of modes form a determinantal point process [4, 5]. Two of them never share a site, and they avoid each other nearby, so their count in a large region fluctuates far less than a random count would: the structure factor falls towards zero at long wavelengths, which is the property called hyperuniformity [6]. Grain made this way looks finer than its crystal count predicts. One layer of tiles shows the difference (figure 1).

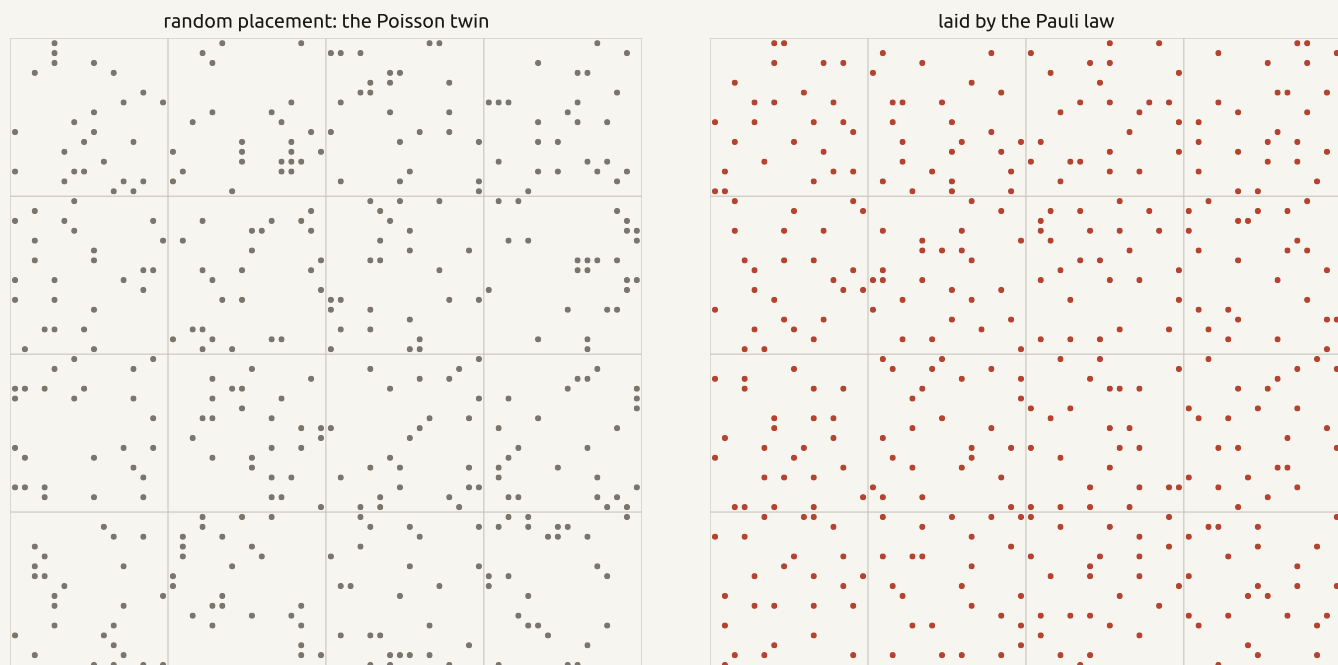


FIGURE 1 One layer of crystals, 4 x 4 tiles of 16 x 16 sites with 25 crystals in every tile. Left, the Poisson twin: the same count, placed at random, clumping and leaving gaps. Right, the Pauli stock: the same count, laid by the fermion law, every crystal keeping its distance. Golden rolls on the streams the first prints use, drawn while this paper was built.

2.1 Two rules for a stock

A stock on this project's shelf obeys two rules, written in the repository before any stock was admitted¹. It must be **physically plausible**: its law is the measurement distribution of a state a circuit can prepare, and it is admitted only with an exact sampler, its authority. Everything after coating - the characteristic curve, the grain's area, the density scale - is borrowed by reference from a real stock in atlas-film, so only the placement of the crystals is new. And it earns the word **unique** only when a film-science statistic measured on it falls outside what a silver halide emulsion gives.

¹ docs/STOCKS.md, "Two rules every stock obeys".

STOCK	ID	TILE	CRYSTALS A TILE	FAMILY	BORROWS
Pauli	pauli	16 x 16	25	determinantal	trix
Pauli, 16-qubit tile	pauli-4x4	4 x 4	5	determinantal	trix
Poisson (reference)	poisson	16 x 16	25	binomial	trix

TABLE 1 The shelf, read off quantum_film/stocks.py while this paper was built. The Poisson stock is the twin every other stock is measured against: the same tile and the same count, placed at random. It is not unique, by design.

3 The law, and the one sampler that decides it

The Pauli stock fills the lowest plane-wave modes of a periodic tile of sites, the modes inside a Fermi disc, with one fermion each, and measures every site. On the shelf's 16 x 16 tile that is 25 crystals a tile; on the 4 x 4 tile a circuit can lay directly, it is 5. Measuring that state lays a projection determinantal process whose kernel is $K(x, y) = (1/M)$ sum over the modes of $\cos(k \cdot (x - y))$. The chance that a site holds a crystal is $K(x, x)$. The chance that two sites both do is $K(x, x) K(y, y) - K(x, y)^2$, never more than the product of their separate chances, and smallest for sites close together.

3.1 The authority

The definition of correct is quantum_film.golden, a sampler in pure Python with mpmath at 256 bits. It draws a roll crystal by crystal by the chain rule [5]: each draw's chances are the diagonal of the kernel conditioned on the crystals already placed, and conditioning on a placed crystal is exactly what digs the hole around it in figure 2. Its uniforms are the first eight bytes of SHA-256 over a named stream, exact rationals that are the same wherever SHA-256 is SHA-256.



FIGURE 2 The exclusion hole. The chance that the next crystal lands on each site of a 16 x 16 Pauli tile, against the average site, with none of a golden roll's crystals placed, then 1, 4 and 12. Each placed crystal digs a dark hole around itself; an average site is pale, and the chance piles up, in the accent, where no crystal has yet landed. The tile wraps at its edges. The stock's own kernel, conditioned on golden roll 1, computed in float64 for the figure.

The only place rounding could choose a crystal is a draw whose target lands near a boundary between two sites' cumulative chances. The authority decides a draw only when its target is at least 2^{224} from every boundary, and refuses it otherwise, by name. Two implementations whose error stays below half that margin decide every draw the same way, so a roll depends on the stock and its seed and on nothing about the machine.

That argument rests on the authority's error being as small as claimed, and the first check of it was worthless: it compared the 256-bit sampler with itself at higher precision, and passed a slip that put every boundary $2^{51.5}$ off. The premise is now measured against code the sampler does not share. On the 4 x 4 tile, and on a 6 x 6 tile that is on no shelf, every cosine the kernel needs is rational and every boundary an exact fraction; on the 16 x 16 stock, an independent chain rule at 512 bits stands in. The 256-bit error in every target and boundary is at most $2^{-249.2^2}$.

2 README.md, "The authority's rolls do not depend on its arithmetic".

4 Sixteen qubits, five fermions

A state of free fermions can be prepared on qubits by a network of Givens rotations between neighbouring qubits on a line [7, 8]. Each rotation mixes the modes of two neighbours and costs two CNOTs and a handful of single-qubit gates. `givens_line` lays the shelf's 4 x 4 law this way, one qubit per site: five qubits start occupied, and 51 rotations in 15 layers spread the five fermions across the line, 102 CNOTs in all (figure 3).

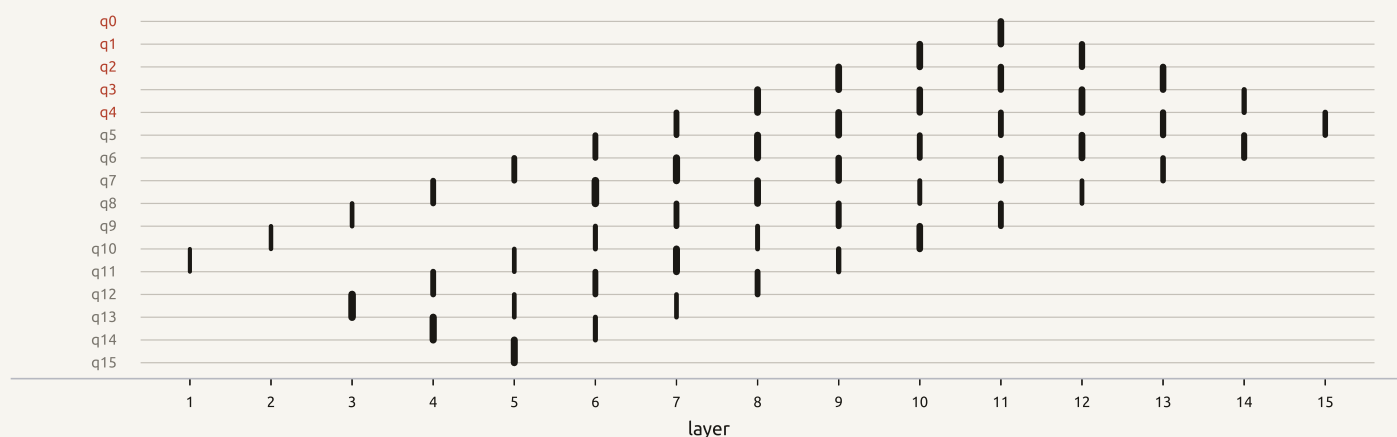


FIGURE 3 The circuit: givens_line’s rotations on the 16-qubit line, layer by layer. Each mark joins the two neighbouring qubits a Givens rotation acts on, and its weight is the rotation’s angle. The first five qubits, in the accent, start occupied. Read from the circuit module’s rotation plan while this paper was built.

Before any of it reached a quantum platform, the circuit was held to the authority on a local statevector: its one- and two-site statistics matched the exact kernel to $4.8\text{e-}15$, and dropping any single rotation broke the match by at least 0.1875^3 . The same statevector check stands in the runner, so the circuit cannot drift from the law without a stage going red.

The circuit’s text is part of its identity. Its QASM was committed beside every run that sent it, with the platform that wrote those bytes, and every score in this paper is stated against that one circuit, whose SHA-256 begins ed767c01bd4b851d.

3 docs/VALIDATION.md, “P3 merged”, 2026-09-26.

5 On Moth's Atlas

Atlas is Moth’s platform for quantum media: engines that run circuits on emulators and, for accounts that have it, on hardware [2]. The account used here had none of the hardware features, so every Atlas run in this paper was on the emulator; the hardware runs of section 10 went to IBM directly. Of the engines, tomography-api-v2 is the one that takes an arbitrary circuit in OpenQASM, and it returns tomography: counts of measured bit strings for each measurement setting, rather than a stream of shots.

Reading those counts carried a trap. Their qubit order changes from one setting to the next within a single response, and nothing in the response says so. Read in plain order, the project’s first run on 25 September reported 11-sigma “disagreements” that were the reader’s, not the physics’⁴. The order was decided against a circuit with a known answer, not against the physics being measured, and every run since has been read only through that one decode rule. Atlas also taught the fixer its first lesson: identical requests returned different bytes wherever sampling was involved, seeded engines included, and a seed fixes the circuit, not the shots.

4 CLAUDE.md, “Atlas: what bites”, item 5.

5.1 Six jobs

The shelf’s Pauli law ran six times, in six jobs of 4,096 shots: the first on 25 September, and the other five on 26 September, spent deliberately before the platform key expired (table 2). Every shot held exactly five crystals.

JOB	SHOTS	DISTINCT LAYOUTS	ONE-SITE MAX Z	PAIRS MAX Z	FORBIDDEN
8586f1cc	4,096	1,995	1.48	2.52	0
579df87d	4,096	1,997	2.53	2.73	0
d017540c	4,096	2,027	1.72	2.14	0
5831ff5e	4,096	2,026	2.33	2.52	0
3673f9fe	4,096	2,006	2.66	2.84	0
933138bc	4,096	2,020	3.14	3.11	0
pooled	24,576	3,004	2.15	2.64	0

TABLE 2 The six runs of the shelf’s 4 x 4 Pauli law on Atlas, each read through the decode rule and scored against the exact law, and the six pooled. Max |z| is the largest deviation, in standard errors, of any one site’s or any one pair’s share of the shots from the law’s chance; forbidden counts shots on layouts the law gives zero chance. Read off the committed records while this paper was built.

Pooled, the 24,576 shots saw 3,004 of the law’s 3,008 allowed layouts and none of its 1,360 forbidden ones. Every one of the 16 sites held a crystal within 2.15 standard errors of the law’s chance, and every one of the 120 pairs within 2.64 (figure 4). One run, 933138bc, is the six’s outlier, with a one-site chi-squared of 34.0 on 16; under the exact law a run that far out has a 5.5% chance among six⁵, and the pooled shots absorb it.

5 docs/VALIDATION.md, the corrections of 2026-09-26, from 20,000 replicates.

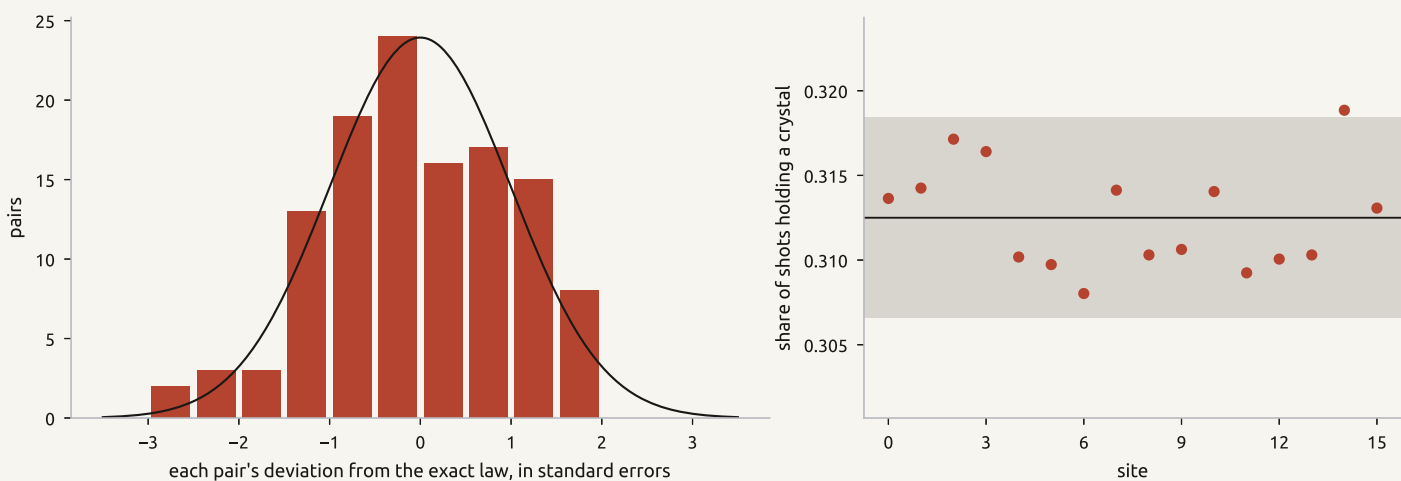


FIGURE 4 Atlas’s 24,576 shots against the exact law. Left: for each of the 120 pairs of sites, how far the share of shots with crystals on both sat from the law’s chance, in standard errors; the curve is what chance alone would give. Right: each site’s share of shots holding a crystal, against the law’s 5/16 and its band of two standard errors. Pooled through the decode rule from the committed results while this paper was built.

Layouts alone cannot tell a coherent state from a mixture that happens to produce the same layouts. The engine's own two-site observables can: in a mixture the off-diagonal correlation of the first pair of sites would be zero, and the state the circuit prepares puts it at 0.375. The first run measured 0.3755 and 0.3687 for the two correlations that carry it⁶, and every run since sits near the same value.

6 docs/VALIDATION.md, "P3 merged", 2026-09-26.

6 Fixed before anyone looked

The fixer is the stage that makes a roll permanent. A record holds the roll's crystals as integers, the law they were laid under and where they came from, and its digest covers all of it. On disk a record is exactly its canonical text, and anything else is refused by name: a changed byte, a duplicated key, a changed law, a missing field. A golden roll is a function of its stock and its seed, so the fixer can re-lay it and compare. A roll from a device cannot be re-laid, so it is recorded and read, never re-run.

A device roll carries one more thing: a commitment over its stock, the circuit's hash, the engine, the job, the number of shots, the decode rule and a salt, committed to the repository before the job's first status call. Before the first status call, and not merely before the result is fetched, because on Atlas a completed job's status response already carries the whole result. What orders the commitment before the result is the code that ran and the local clock; nothing third-party dates the commit, and the record says so rather than claim more.

The six runs left 12,071 device rolls, one for each distinct layout in each job, with the number of shots it occurred in, and the fixer's own command line accepts every one. They are the material of the first print laid by Atlas, and the reason that print can be re-developed: its record names the jobs, the order in which their shots were laid, and the digest of every stage after.

7 Equal by construction

The authority is exact and slow. A second sampler was built to lay the same rolls in ordinary binary64 on libcft, the pinned arithmetic of cft-fp256 [9], and to lay them identically, not approximately. It never trusts binary64 to be close. Every chance, boundary and target is bracketed by directed rounding around its exact value; a draw is decided only when the target's bracket clears every boundary by 2^{-200} , which is 2^{24} times the authority's own margin; and a roll with any draw it cannot decide goes whole to the authority. On 2,300 pauli-4x4 rolls and 740 pauli rolls it laid the authority's roll every time, and none was handed off. On the 16 x 16 stock it takes 0.62x the authority's time⁷.

7 docs/VALIDATION.md, "P1 merged", 2026-09-26.

Equality with the authority turned out to be the wrong test. 10 of 11 faults planted in the bracketing code passed it, because a bracket that is too loose still decides the same draws on random inputs. What catches them is holding each step to exact rationals on inputs tight enough that only that step's own roundings separate its bound from the exact value, and flipping every directed rounding in turn to watch its step fail. A later fault taught the same lesson one level up: an end-to-end check that read what the sampler traced, rather than the decision it returned, passed a fault that traced honestly and returned another answer. The check now reads the decision.

8 From rolls to film

Everything after coating is borrowed from atlas-film, the film model this project develops through [10], by reference to a real stock: here TRI-X, as atlas-film models it. atlas-film's negative is a sheet. Each pixel-cell holds some number K of crystals, each with a fixed 16-bit sensitivity threshold, and development is a comparison, not a draw: a crystal develops if its threshold lies at or below the fraction of crystals the exposure renders developable.

A quantum roll lays at most one crystal a site, and TRI-X, even at its finest honest pitch, coats about nine crystals in every column of emulsion. So a site cannot be a cell's only crystal, and the emulsion's depth holds layers instead. Each layer is one roll of every tile; each layer's grid is shifted by an offset drawn from the print's stream, so the tiles' seams do not line up through the depth; and a cell's K is its column's count over the layers. Each crystal's threshold is the top 16 bits of the same SHA-256 uniform the authority uses, on the print's stream.

Because every roll holds exactly its stock's count of crystals, the sheet's mean K is fixed by the number of layers, and the pitch is derived from it so that the borrowed stock's own coating density equals that mean exactly - a check atlas-film's own sheet path does not make. The pitch must also clear the stock's honesty floor, a cell at least one crystal across: at TRI-X's density the 16 x 16 Pauli stock needs at least 91 layers, and the 4 x 4 stock at least 29.

The sheet is integers, laid by integer arithmetic and SHA-256. It is developed and printed in atlas-film's pinned mode, whose negative and print are the same bits on every IEEE 754 binary64 machine, under gates and stated limits of its own; a trace hook installed at import, for one, can still rewrite a chain from outside, and its documentation says so⁸. The prints are made on a silver paper without grain of its own, so the grain on them is the film's.

⁸ atlas-film, docs/PINNED.md, on the branch pinned.

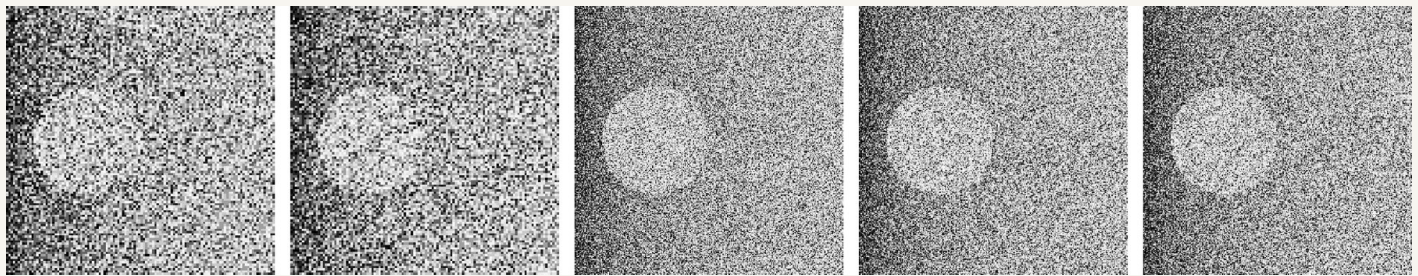


FIGURE 5 The five first prints, as the repository holds them, each pixel one column of crystals. From left: the 4 x 4 Pauli law laid entirely by Atlas; the same law from its exact sampler; the 16 x 16 Pauli stock; its Poisson twin; TRI-X as atlas-film coats it. All five develop the same scene, a ramp with a bright disc; the two 4 x 4 prints share one pitch and density, and the other three another.

PRINT	COLUMNS	LAYERS	PITCH, UM	MEAN K	PRINT DIGEST
atlas-pauli-4x4	116 x 116	29	1.2526	9.0625	78c70404809c0da3
golden-pauli-4x4	116 x 116	29	1.2526	9.0625	971a5446da0594a1
pauli	256 x 256	91	1.2404	8.8867	96415fe7964c0a6c
poisson	256 x 256	91	1.2404	8.8867	bd468d75946b56b9
trix	256 x 256	91	1.2404	8.8734	e6b6d9292142dd31

TABLE 3 The five first prints, read off their records in docs/prints/. Each record also names its rolls, its stream, its scene's digest and both exposures as exact binary64 values; re-developing a print from its record alone gives the same bits.

The first print holds 24,389 of the 24,576 Atlas shots, stacked 29 deep in 29 x 29 tiles, in an order drawn from its own stream: every crystal in it was laid by Atlas. The 187 shots left over are counted in its record as unused.

9 What the grain shows

To the eye, at one column of crystals a pixel, the prints look alike. The difference is in the structure of the grain, and the structure factor of each sheet's crystal count shows it (figure 6). The numbers are read between a tile's scale and the Fermi scale - on the 256-column sheets, from 16 to 45 cycles across the sheet - because that is where the law, rather than the tiling, decides the grain.

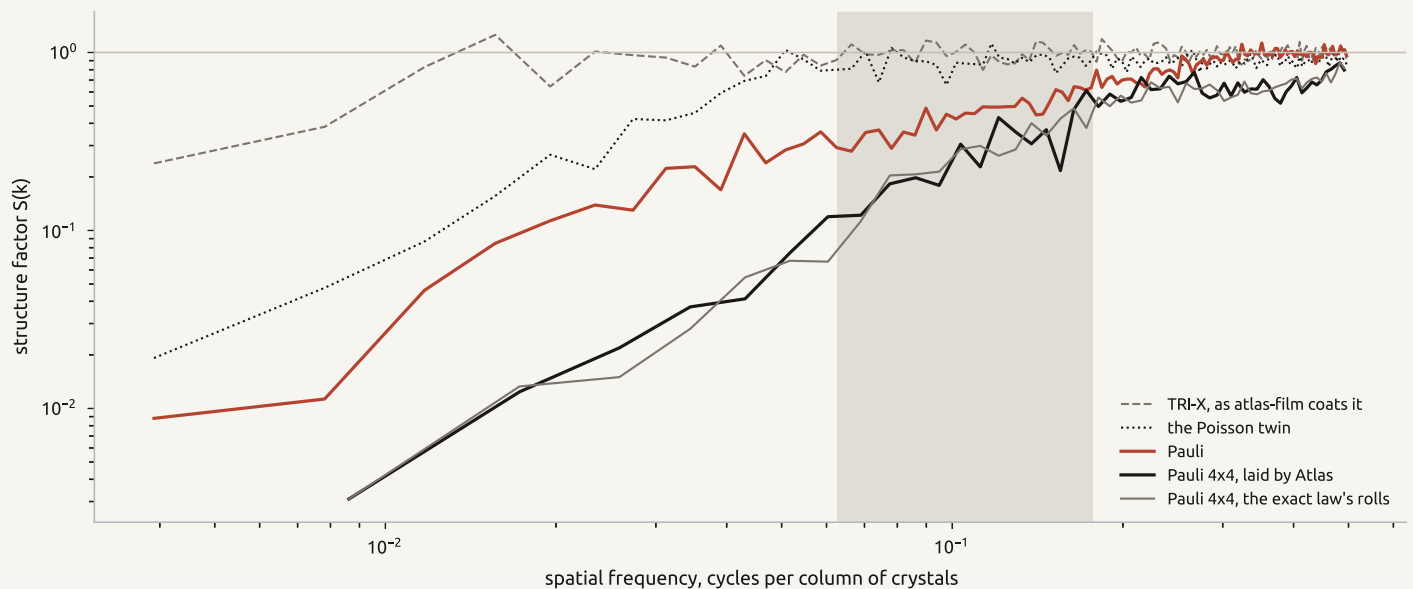


FIGURE 6 The structure factor of each print's crystal count, read off the prints' records. The shaded band runs from the Pauli tile's scale to its Fermi scale. TRI-X, coated by atlas-film at random, sits near 1, its lowest bins scattering because they hold few modes; the Poisson twin, the same count per tile placed at random, falls only below the tile's scale; Pauli sits well below both across the band. The 4 x 4 prints, laid by Atlas and by the exact law, trace one curve.

In that band the Pauli sheet's crystal count carries between 0.279 and 0.642 of film's noise power, 0.472 on average. The Poisson twin averages 0.892 there and TRI-X 0.997, and every Pauli value in the band lies below every value of the twin. The exact law expects 0.301-0.644 for Pauli⁹, so the sheet is what the law says it should be. The 4 x 4 prints trace one curve: at 8, 16 and 32 cycles the Atlas sheet reads 0.122, 0.307 and 0.592, and its twin from the exact law 0.113, 0.401 and 0.628.

9 docs/VALIDATION.md, the corrections of 2026-09-26.

9.1 What the tiling does, and what it does not

A sheet built from tiles that each hold a fixed count of crystals has a property of its own. The fixed count sends the structure factor to zero at the longest wavelengths for every law, and for crystals placed at random that suppression reaches up to about a tile's scale, which is why the twin, not atlas-film's free coating, is the fair comparison below it. It is not a property of every law. A verifier's control on the prints' own geometry, with crystals bunched at a random place in each tile, came out enhanced at those scales, not suppressed: 1.6 to 11.7 where the twin sits below 1. The project first wrote the opposite, that tiling suppresses the long wavelengths for any law; the verifier found it false, and the correction stands in the ledger beside the sentence it corrects. It will matter for the first stock whose crystals bunch.

10 On IBM's quantum computers

10 docs/PREREGISTRATION.md.

The Atlas account had no hardware, so the circuit went to IBM Quantum directly, on its free Open plan: ten minutes of processor time in every 28 days, on three Heron r2 processors of 156 qubits, `ibm_kingston`, `ibm_marrakesh` and `ibm_fez` [3]. Nothing that samples on hardware repeats, so everything a result would be held to was fixed before the first job ran. The circuit was compiled once for each device and frozen beside its hash. The measures, and what each device's own noise model predicted for them, were written into a pre-registration and pushed¹⁰. And each job's line, naming its bundle and its circuit, was committed and pushed to GitHub after the job was submitted and before its result was read.

Each device ran a circuit whose answer is known in advance first, to settle the order in which its bits are read, the trap Atlas had set. All 31 known-answer jobs held, and no shot read the mirror order. In all the three devices ran 65 jobs, and the shelf's law ran 761,856 times.

10.1 What noise did to the law

The law forbids 1,360 of the 4,368 ways to place five crystals on the tile, and Atlas's emulator laid none of them. Noise breaks a shot in two ways that can be counted (table 4). A shot can gain or lose a crystal: in each device's first law job, only 52% to 61% of the shots kept exactly five. And a shot that keeps five can land where the law forbids: 4.3% to 5.4% of them did, where crystals placed at random land there 31.1% of the time. The two correlations that carry the state's coherence survived, decayed: between +0.269 and +0.344 on the devices, against the law's +0.375 and the zero that a classical mixture of the same layouts gives. The linear cross-entropy benchmark, which scores the law 1 and random placement 0, came out between +0.806 and +0.849.

Every device fell short of its own noise model. The models predicted a benchmark of +0.921 to +0.939, and on each device the forbidden share came out 1.92 to 2.08 times the prediction. The models leave out crosstalk, leakage and drift, and the pre-registration said so before any job ran.

	FIVE CRYSTALS	FORBIDDEN, OF THOSE	LINEAR XEB	XX	YY
the exact law	100.0%	0.0%	+1.000	+0.375	+0.375
Atlas's emulator	100.0%	0.0%	+1.012	+0.377	+0.369
random placement	100.0%	31.1%	+0.000	+0.000	+0.000
<code>ibm_kingston</code> : its noise model	72.2%	2.2%	+0.939	+0.363	+0.357
<code>ibm_kingston</code> : its first jobs	60.9%	4.3%	+0.844	+0.332	+0.330
<code>ibm_kingston</code> : 9 law jobs pooled	59.6%	4.3%	+0.857	-	-
<code>ibm_marrakesh</code> : its noise model	69.3%	2.5%	+0.921	+0.363	+0.353

	FIVE CRYSTALS	FORBIDDEN, OF THOSE	LINEAR XEB	XX	YY
ibm_marrakesh: its first jobs	51.9%	5.1%	+0.849	+0.313	+0.344
ibm_marrakesh: 11 law jobs pooled	52.7%	5.2%	+0.843	-	-
ibm_fez: its noise model	67.7%	2.7%	+0.921	+0.356	+0.361
ibm_fez: its first jobs	53.1%	5.4%	+0.806	+0.269	+0.305
ibm_fez: 11 law jobs pooled	54.4%	5.1%	+0.841	-	-

TABLE 4 The pre-registered measures, read off the committed IBM runs by the project's comparison while this paper was built. Five crystals: the share of shots that kept exactly five. Forbidden: the share of those that landed on a layout the law gives zero chance. XX and YY: the two correlations of the first pair of sites that carry the state's coherence, zero for a classical mixture of the same layouts. For each device: what its noise model predicted, its first jobs, and all of its law jobs pooled; the later jobs measured no coherence. The emulator's row is Atlas's 24,576 shots.

10.2 A sheet from each device

Each device then ran more of the same jobs, 56 in all, so that its five-crystal shots would lay a sheet as large as the shelf's, by the print rule the pre-registration fixed: 268, 280 and 284 columns, stacked 29 deep (figure 7). To the eye they are the exact law's sheet. Pooled over each device's law jobs, 4.3% to 5.2% of the five-crystal shots fell on forbidden layouts.

The repeated jobs also caught the devices drifting. Each device's five-crystal share moved between jobs by 9 to 13 times one job's sampling error, and on ibm_marrakesh it stepped by +0.0306 between two jobs submitted 47 seconds apart. What changed on the device is not recorded.

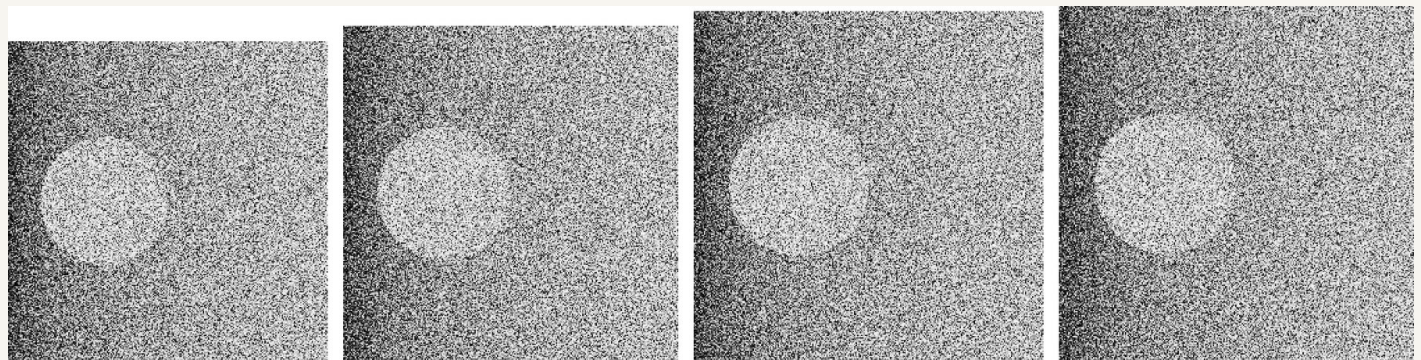


FIGURE 7 The exact law's sheet and the three IBM sheets, at one scale, each pixel one column of crystals. From left: the exact law's own rolls; ibm_kingston; ibm_marrakesh; ibm_fez. Each IBM sheet is its device's five-crystal shots, laid by the pre-registered print rule and developed like every other print here. Read off docs/prints while this paper was built.

10.3 The forbidden shots, in colour

A print sums 29 layers and hides which of its shots were forbidden. So each sheet's shots were laid again exactly as its print laid them, the print re-developed and held to its

recorded digest, and every crystal of a forbidden layout tinted (figure 8). This part was not pre-registered: it was made afterwards, at Logan's suggestion, and it is exploratory.

The law forbids clumps. A shot laid by the law holds exactly 2 neighbouring pairs among its five crystals on average, and never more than 4; the layouts it forbids hold 3.11 on average. So when noise breaks exclusion, what it lays is clumps. The hardware's forbidden shots held 2.91 to 3.01 neighbouring pairs on average, and its allowed shots 2.03 to 2.05: the grain that Pauli exclusion takes out of an emulsion, put back by the machine's noise.

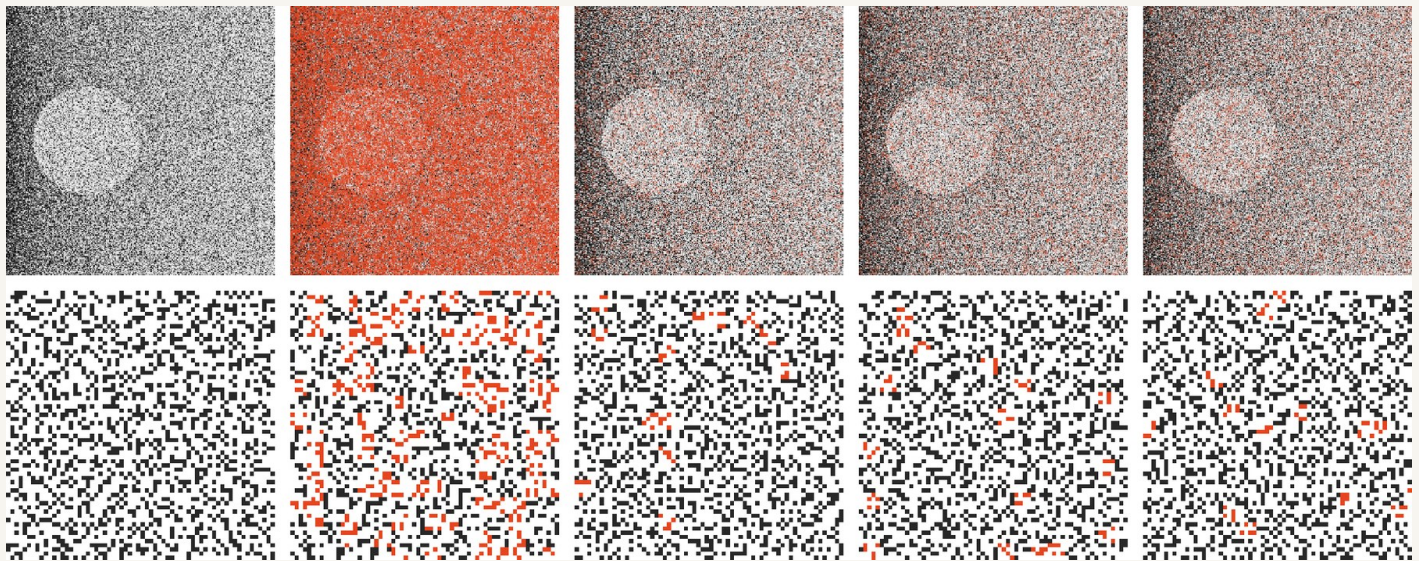


FIGURE 8 The layouts the law forbids, in colour. Top: each sheet, every column of crystals tinted by twice the share of its crystals that came from forbidden layouts. Bottom: a corner of each sheet's first layer, 64 columns on a side, shot by shot, every crystal of a forbidden layout in the tint. From left: the exact law, which lays none; random placement, the twin; `ibm_kingston`; `ibm_marrakesh`; `ibm_fez`. Re-laid from the records and held to each print's digest when the repository drew them; read off docs/prints while this paper was built.

10.4 What this does not show

The runs are one day on three processors of one maker, with error mitigation off by design and no processor time bought. They measure how far each device was from the law and from random placement that day. They do not say how a device varies from day to day, and they pass or fail nothing.

11 How it is checked

One definition of correct. The authority is the only judge. The circuit, the `binary64` sampler, the `develop` seam, and Atlas's and IBM's results are all scored against it, and never against each other. A second opinion that disagrees with the authority is a bug report, not a vote.

One command. The runner, `verify/run.sh`, has 18 stages: lint, the documents, the frozen vectors, the authority, the circuits, the decode rules, the Atlas client offline, the fixer, the IBM runs and their records without qiskit, the binary64 sampler, the develop seam, the comparison, the noise models' predictions, the fixer's own command line, a negative control and its twin, the IBM path with qiskit but offline, libcft, and the live Atlas account. A stage that cannot run skips by name, never silently, and a test that skips inside a stage fails the stage, because a skip is not a pass. The negative control feeds the fixer's command line a record with one crystal moved, which it must refuse; the twin feeds it the genuine records, which it must accept, because a command line that refused everything would pass the control alone.

Every gate watched to fail. A gate is trusted green only after a planted fault has turned it red: a moved crystal, a shrunken margin, a rounding in the wrong direction, a shuffle that returns the wrong permutation. Several gates did not survive that test when they were first written. The first check of the authority's margin compared it with itself; a rule against binary64 in the authority read spellings, and a respelling walked past it; a test of the pinned sampler read what it traced instead of what it decided. Each was replaced by a check on behaviour, and each replacement was watched to fail too.

Verifiers, not reviewers. Each parcel of the work, and the lead's own, was checked by an independent agent whose brief was to disconfirm it and never to fix it, working in its own clone, against a stated bar: a correctness defect, a claim false as stated, or a fault that passes every gate outside every stated limit. Where a gate could not be written, the limit was stated instead, and a verifier's next job was to test whether the limit was stated truly. In the hardware round the verifiers were Claude Sonnet 5.5 models, at Logan's word, as a test of that family at this work. The web demo and the poster, made last as presentation, had no verifier; a test holds their data to the records instead.

A ledger that keeps its mistakes. The validation ledger, `docs/VALIDATION.md`, is append-only. Each entry is a run: what it measured, and what it got wrong. A later correction is a later entry, beside the sentence it corrects, never an edit of it. It holds corrections of the project's own prose: two labels copied from a report without their definitions, the two sentences about the structure factor that section 9 now states correctly, and, in the hardware round, a correlation given as +0.314 that was a second rounding of 0.3135.

12 What it is not, and what comes next

It is not a quantum advantage: at these sizes both laws are simulated exactly on a laptop, which is what lets every shot be checked. Its hardware is one day on three processors of one maker, which says nothing of how they vary from day to day. It is not a model of any real film's grain; the Poisson placement is the reference, and Pauli is the point. It is not yet a colour film, or a whole frame. And a device roll's commitment is anchored by the code

and the local clock, not by anything a third party has dated, though each IBM job's line was also pushed to GitHub before its result was read.

12.1 A whole sheet

A frame of 35 mm film is a sheet 36 by 24 millimetres. At TRI-X's density, as atlas-film models it, every square millimetre holds about 5.8 million crystals¹¹, so a frame holds about 5 billion, about 200 million rolls of the 16 x 16 Pauli tile. That is an estimate from the model's constants, and it sets the next piece of work: a certified sampler fast enough for a frame, batching many tiles through each call to the pinned arithmetic, and, on hardware, circuits that lay whole tiles in one shot. The first prints are small because the samplers are, not because the method is.

11 docs/ROADMAP.md, P2, "Still to come".

12.2 Colour

atlas-film coats a colour film as three layers. A Pauli tripack would place its crystals with one fermionic state across site and layer together, so that a crystal in one dye layer repels a crystal in the layer beneath it as well as its neighbours in its own: colour grain whose layers are anticorrelated, which no colour negative has¹². Two things stand between it and a print: atlas-film's colour chain has to accept three supplied sheets, a small change upstream, and its colour path has to be pinned, which pinned mode refuses today.

12 docs/STOCKS.md, "The unique-stock programme".

12.3 A quantum hand on the response

Today a quantum measurement decides where each crystal lies, and a hash decides how readily it responds to light: each crystal's sensitivity threshold is a uniform drawn from SHA-256, and the characteristic curve is those thresholds read against exposure. The next step is to let the quantum state decide the response as well - thresholds drawn from a measured distribution, or tied between neighbouring crystals by an entangled state - so that the curve itself, its steps and its noise, carries the physics and not only the placement. Nothing of this is built. It is a direction, and it will need an authority of its own before a stock can use it.

12.4 Bunching, and noise

Speckle is the planned stock with the opposite character: Born-rule shots of a random pupil through a quantum Fourier transform, crystals that bunch rather than repel, and a grain contrast equal to the fidelity of the machine that exposed it. It would be the first stock whose grain measures the machine that made it, and its twin will have to be measured rather than assumed, because a bunched law is exactly the case in which tiling enhances the long wavelengths instead of suppressing them. The shelf's Pauli law has now met noise. Before any processor ran it, the project named a device's fraction of forbidden layouts its first witness of noise¹³; on IBM's processors it came out at 4.3% to 5.2%, where the emulator's 24,576 shots put it at zero. Moth offered to run a circuit for the project, and the frozen circuit and these results are packaged for it¹⁴; running it is Moth's choice.

13 docs/DETERMINISM.md, "Evidence, and its gaps".

14 docs/MOTH-HANDOFF.md.

13 How it was made

All of this was built for Moth Hack 2026: the first two rounds in two days, 25 and 26 September 2026, and the third, which took the circuit to hardware, on 3 and 5 October. The repository's first commit is dated 25 September 2026 at 11:56, local time, and it reached the state this paper describes in 230 commits (figure 9). Every one of those commits carries a co-author line naming a Claude model.

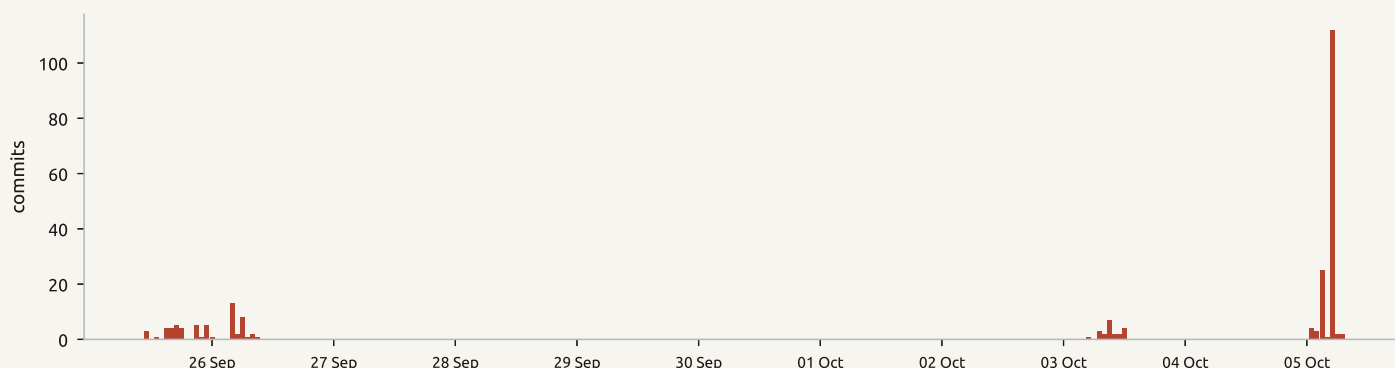


FIGURE 9 Commits per hour, local time, from the repository's first commit to the one this paper describes. The tall bar on 5 October is the sheets' run: the run script committed each IBM job's line before its result was read, and the result after it. Read from the git log while this paper was built.

The work was split the way ParcelRound splits work [11]. A lead, one Claude Opus 5.5 session, built the seams first: the shelf, the authority, the fixer, the Atlas client and the runner. Logan asked for one more step before anything was handed out: an independent verifier on the lead's own work, before any parcel was dispatched. Three parcels then ran in parallel, each an agent with a written brief naming the files it owned and the files it must not touch: the binary64 sampler, the pinned mode of atlas-film in atlas-film's own repository, and the hardware-shaped circuit with its runs on Atlas. Each had its own verifier, whose brief was to disconfirm and never to fix, and a parcel merged only on its verifier's word. The round ran overnight, while Logan slept, with the lead working to a standing instruction and a ledger outside every repository as the round's shared record. The second round, the develop seam and the first prints, was the lead's alone, with one verifier pass at its end. The third took the circuit to hardware. The lead built the record a hardware run leaves, its commitment and IBM's bit reader, and had them verified before anything was dispatched; then two parcels ran, the frozen bundle with its runner and the comparison with its predictions, each with its own verifier.

Logan decided what mattered: that a verifier would check the lead before the lead checked anyone; that the platform key's last five jobs would be spent on a print laid entirely by Atlas; that atlas-film would be vendored at its pinned branch; that the second round would be fast. In the third, Logan decided that IBM's free plan would come first, where a mistake is cheap; that no hardware job would run without Logan's word; that the verifiers would be Sonnet models; that the plan's remaining minutes would buy a sheet

from each device; and that the forbidden shots should be shown in colour. Claude Opus 5.5, and the agents it directed, wrote most of the code and the documents, this paper included. The project counts them as equal contributors, and so the paper carries both names; for the same reason, and by its owner's intent, the Moth Hack entry is not a candidate for cash prizes.

None of that is what makes the results trustworthy. What does is the discipline the repository enforces on everyone who works in it, human or not, and the previous sections describe it: one authority, gates watched to fail, verifiers that disconfirm, and a ledger that keeps its mistakes. The first two rounds' own record, with their timeline, their defects and what the method should learn from them, is ParcelRound's fourth case study; the third round kept a ledger of its own, outside every repository.

How this document was made

Nothing in this paper was typed where it could be read. The sections are Markdown; each figure is a maker in a script beside them that reads Quantum-Film at build time - the golden samplers and the stock table, the circuit's rotation plan, the committed Atlas and IBM results through their decode rules and the project's comparison, the prints and their records, and the git log - and writes a sidecar beside the file with the numbers it drew and the commit it read them at. The tables ask the same script for their rows. The numbers the prose states are recomputed by a check that fails the build if a sentence and its source disagree, or if the sentence has quietly stopped being there; a measurement that cannot be recomputed is cited instead, and the check confirms that the document named beside it says the same.

The type is Ubuntu and Ubuntu Mono, under the Ubuntu Font Licence. The page is a letter sheet with a column in the outer margin for the notes, and the PDF is printed from the same HTML the editor shows. It was built with storydocs, the pipeline the atlas family's papers were made with.

Sources

- ¹ White, L. and Claude Opus 5.5 (2026). *Quantum-Film*. source repository, MIT licence. <https://github.com/loganw234/Quantum-Film> everything this paper describes; its web demo is at <https://loganw234.github.io/Quantum-Film/>
- ² Moth Quantum (2026). *Atlas*. quantum media platform and API. <https://platform.mothquantum.com> the tomography-api-v2 engine, on the emulator
- ³ IBM Quantum (2026). *IBM Quantum Platform*. quantum computing service, Open plan. <https://quantum.cloud.ibm.com> ibm_kingston, ibm_marrakesh and ibm_fez, Heron r2 processors, reached through Qiskit Runtime's sampler

- 4 Macchi, O. (1975). *The coincidence approach to stochastic point processes*. Advances in Applied Probability, vol. 7, no. 1, pp. 83-122. [10.2307/1425855](#) the fermion point process, the law the Pauli stock samples
- 5 Hough, J. B., Krishnapur, M., Peres, Y. and Virág, B. (2006). *Determinantal processes and independence*. Probability Surveys, vol. 3, pp. 206-229. [10.1214/154957806000000078](#) the chain rule for a projection kernel, the sampler the authority is
- 6 Torquato, S. and Stillinger, F. H. (2003). *Local density fluctuations, hyperuniformity, and order metrics*. Physical Review E, vol. 68, pp. 16971. [10.1103/PhysRevE.68.041113](#) hyperuniformity, a structure factor that vanishes at long wavelengths
- 7 Kivlichan, I. D., McClean, J., Wiebe, N., Gidney, C., Aspuru-Guzik, A., Chan, G. K.-L. and Babbush, R. (2018). *Quantum simulation of electronic structure with linear depth and connectivity*. Physical Review Letters, vol. 120, pp. 110501. [10.1103/PhysRevLett.120.110501](#) Givens rotation networks on a line of qubits
- 8 Jiang, Z., Sung, K. J., Kechedzhi, K., Smelyanskiy, V. N. and Boixo, S. (2018). *Quantum algorithms to simulate many-body physics of correlated fermions*. Physical Review Applied, vol. 9, pp. 18462. [10.1103/PhysRevApplied.9.044036](#) preparing a Slater determinant with Givens rotations
- 9 White, L. (2026). *cft-fp256*. source repository, Apache-2.0 licence. <https://github.com/loganw234/cft-fp256> the pinned arithmetic, libcft
- 10 White, L. (2026). *atlas-film*. source repository, MIT licence. <https://github.com/loganw234/atlas-film> the film model; its pinned mode is on the branch pinned
- 11 White, L. (2026). *ParcelRound*. source repository. <https://github.com/loganw234/ParcelRound> the method the work was split by; this project's first two rounds are its fourth case study